Do Now · retrieval

Computational problem recognition

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why may Spiral be preferable for an uncertain high-risk project?

Q2

What is abstraction?

Q3

What is a base case?

Q4

For C# array int[] x={2,4,6,8}, a loop doubles elements at even indexes (0,2). Give the final array and sum.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Recognise ambiguity — 'Fair' is subjective until objective criteria are supplied, such as balancing first-period lessons across teachers.

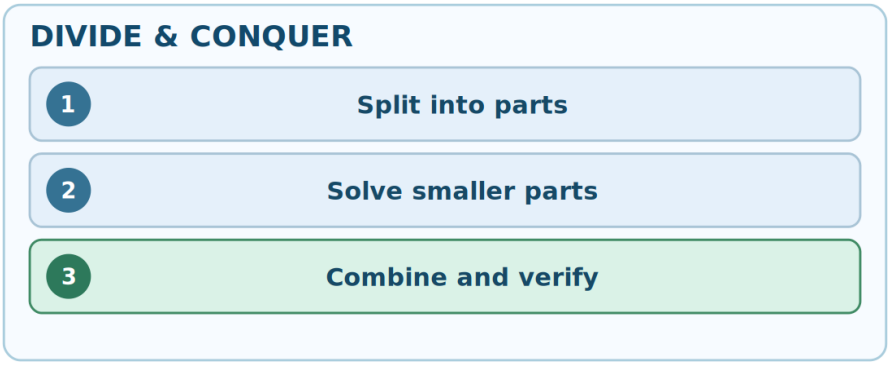
Core idea 2

Make requirements testable — Represent rooms, lessons and slots; specify hard constraints and a success test.

Core idea 3

Select techniques appropriately — Decompose into assign/check/evaluate modules. Divide and conquer is useful where subproblems can be solved and combined consistently; not every decomposition is divide and conquer.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

problem recognition — Determine what problem is being solved and its computational form.

decomposition — Split into manageable subproblems.

divide and conquer — Divide input, solve smaller instances, combine results.

constraint — Condition a valid solution must satisfy.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

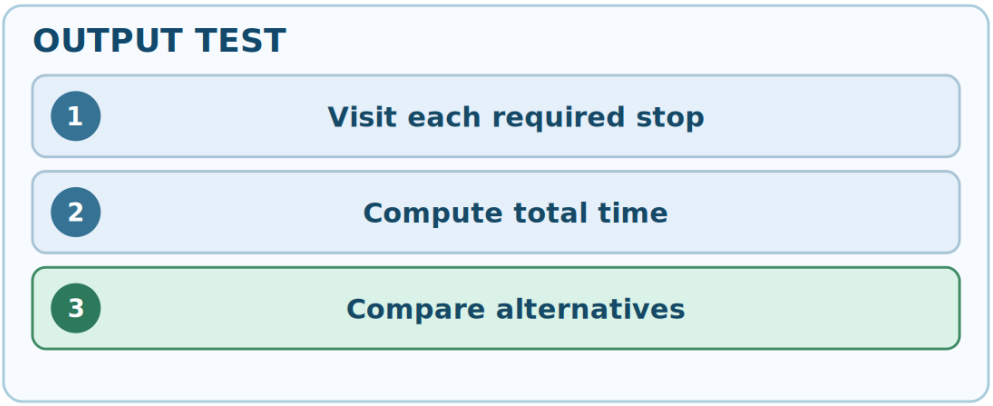
Rewrite 'choose the best delivery van route' into a computational task.

Follow the worked method

1. Predict the method — Replace vague 'best' with a measurable objective such as minimum total travel time.

2. Check and explain — Specify route constraints and verify a produced route covers all required locations.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict the method
Use explicit demand and stock data and define what fairness means with measurable priorities.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

For a system allocating parking spaces, give one input, one constraint, one output and one test of correctness.

Question 2

Explain why 'select the nicest library books' is not computable as stated; propose precise inputs, a ranking rule and an output.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare decomposition and divide-and-conquer for sorting 1,000 items, using merge sort as an example.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An unclear human judgement becomes computable simply by choosing a faster processor.

Use key vocabulary from memory

1. Explain problem recognition in your own words.

2. Explain decomposition in your own words.

3. Explain divide and conquer in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.